



Research on Experience Design of Social Reading Platforms Based on Social Presence Theory

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Abstract. With the development of the Internet, digital reading is becoming more and more popular. The change of reading media strongly affected users' reading behavior. In the Era of Web 3.0, the sociality of reading has been amplified by the Internet. A social reading model with readers as the core and emphasis on sharing, interaction and dissemination came into being. This paper introduces social Presence theory to study how to enhance the perception among users on platforms, thereby improving the user experience of social reading. Firstly, through literature research, we subdivided the social presence on social reading platforms into conscious social presence, emotional social presence, and cognitive social presence. We extracted the design elements of social reading platforms which might lead to high social presence then. By card sorting, design elements were grouped. Finally, we proposed an experience model of social reading platforms which explains the relationship between social presence and user experience as a reference for the interaction design of social reading platforms in the future.

Keywords: Social Presence · Social reading · User experience

1 Research Background

With the popularity of mobile Internet, people's reading behavior has changed dramatically. From paper books to Kindle ink screens to digital reading, people have more choices. Digital reading is welcomed by users because of its high degree of personalization, rich content, and other advantages. In 2020, the contact rate of digital reading (online reading, mobile phone reading, e-reader reading, iPad reading, etc.) among Chinese adults was 79.4%. Digital reading has become an important way for many readers to obtain information and knowledge. [1] The number of digital readers in China is growing, and various media are integrating and complementing each other. Reading is entertainment, and reading and social interaction are deeply integrated.

In the Era of Web 3.0, the sociality of reading is amplified by the Internet and various social reading platforms. With readers as the core, the social reading model that emphasizes sharing, interaction and dissemination arises at the historic moment. However, in digital reading, social communication transmits information through digital media.

Compared with reading discussion, information and emotion between users inevitably suffer certain loss in transmission, thus affecting user experience.

At present, China's research on social reading mainly focuses on social reading theory, social reading intention, social reading behavior and social reading practice. However, the research on user experience in social reading is insufficient, so this paper focuses on the experience design of online reading community. This paper discusses how to reduce the sense of distance brought by media through design and make interactive experience more vivid, to improve users' social reading experience.

2 Social Presence and User Experience in Reading Communities

2.1 Social Presence Theory

The theory of social presence was first proposed by Short et al. in 1976. They believe that social presence refers to the degree to which a person is regarded as a "real person" and the perceived degree of connection with others in the process of communication through media, which is a media attribute [2]. Tu believes that social presence occurs in the media environment and relates to people's awareness of others [6]. Social presence is not only supported by media, but also a psychological perception, which is affected by both technical and social factors [3].

Social presence was first proposed and applied in the field of communication. With the development of the Internet, the boundaries of various fields are constantly broken, and integrated with each other. Therefore, social presence theory has been gradually applied to more fields related to communication and social interaction, such as online learning, telemedicine and so on. Wang Han et al. applied social presence to the research of digital reading, and explored the influencing factors and causes of users' pan-recreational reading [4]. Ji Dan et al. proposed through research that social presence mediated by satisfaction has a positive impact on users' reading behavior [5]. In a study of user experience in commercial communities, Wang divides social presence into three types: cognitive social presence, emotional social presence and conscious social presence, which provides references for the study of this paper [6].

2.2 User Experience

In 1982, Steve Jobs and Hartmut Eslinger established the value of "user-centered, design-centered" for Apple. Don Norman and Jakob Nielsen firstly put forward that user experience encompasses all aspects of the end-user's interaction with the company, its services, and its products [7]. The core of user experience design is to consider the overall user experience from the user's point of view. Typical methods of user experience design include user journey map, card taxonomy, etc. With the development of Internet products, more and more attention has been paid to user experience. Good user experience can bring users a pleasant feeling, thus improving satisfaction and turning ordinary users into loyal users.

2.3 The Relationship Between Social Presence and Reading Community Experience

Reading, as a kind of dialogue, first takes place between individuals and texts, and then extends to texts and readers. The reading experience is inherently social. Personal interpretations are shaped and consolidated by constant discussion with other readers. The social reading mode supported by the reading community is a new digital reading mode with readers as the center and social relations as the link. Social reading mode emphasizes sharing, interaction and communication, pays more attention to user experience advocates user creation of content, attaches importance to joint communication and profit, and realizes infinite amplification of reading value on the basis of multi-directional interaction.

Compared with the traditional reading mode, the reading experience generated in the process of social reading is more complex and diverse. Li et al. summarized the reading experience under the new technology environment into aesthetic experience, emotional experience, thinking experience and other traditional reading experience, value experience, social experience, immersion experience and other dimensions. Moreover, from the perspective of experience level, value experience, social experience and thinking experience are the development products of aesthetic experience and emotional experience as general experience. On this basis, immersion experience is finally achieved, namely the peak of experience [8]. Currently, there are few studies on the specific formation patterns and design strategies of user experience in online reading communities under new technologies and media.

Social presence theory mainly studies the cognitive feelings of users when they communicate and interact with others through the Internet and other media. Many studies have shown that social presence has a positive impact on immersive experience, useful perception, etc. One of the biggest differences between a reading community and a traditional reading or purely digital document reading tool is the emphasis on the interaction of multiple readers. The impact of socializing through reading between users on the experience is a key part of user experience research in reading communities. It can be concluded that the two research scenarios match each other. Therefore, this paper analyzes the influence of design elements in reading community on user experience from the perspective of social presence, and then comes to the experience design strategy of online reading community.

3 Read Community Experience Elements from the Perspective of Social Presence

3.1 Extraction of User Experience Elements of Social Reading

In order to study the influence mechanism of social presence and social reading experience, this paper subdivides users' social presence in online reading community into conscious social presence, emotional social presence, and cognitive social presence, and respectively explores the design elements. To ensure that the design elements in the design model are verified in real design practice, the discussion takes "WeRead", a typical online reading community, as a case study.

“WeRead” is a reading software based on WeChat relationship chain which is popular in China. The slogan “make reading no longer lonely” reflects its product positioning for social reading. Through the operation of the reading community, WeRead promotes the interaction between readers, thus creating a warm and not lonely reading experience.

We recruited 8 WeRead users as interview subjects. We discussed and recorded their social reading experience and related experience on the platform. And pull out the relevant experience elements Including user profile pictures, comments, likes and so on.

Then, three users were invited to classify the elements together with us through the card sorting. All the elements were divided into three groups, and the following table was obtained (see Table 1).

Table 1. Design elements on social reading platforms

Original elements	Element groups	Social presence
Avatar, nickname, gender, region tag, personality signature.....	Users’ identity elements	Conscious social presence
Comments, likes, Dynamic effect.....	Reading interaction elements	Emotional social presence
Graphic links, context, Browsing path.....	Reading situation elements	Cognitive social presence

3.2 Conscious Social Presence and Reading Community Experience Elements

Conscious social presence refers to that readers are aware of other social participants’ virtual presence and feel their social influence in the process of reading community experience. Good design can make users aware of the real existence of the users behind them when facing other user accounts in the community, instead of a bunch of cold digital information or robots.

Take WeRead as an example, there will be other users’ identities exposed before, during and after reading. The display of user identity mainly includes avatar, nickname, gender, region tag and personality signature. WeRead also transfers the original social relations in WeChat and sets the label “WeChat friends”, which greatly reduces the cognitive cost of friend identity. This was followed by the number of hours spent reading, likes and followers. Reading duration can reflect the richness of reading experience and indirectly reflect reading thinking ability. The number of likes and followers can reflect her reading appreciation ability through social influence. Meanwhile, it can reflect social influence, arouse the group consciousness of users, and attract users’ attention more.

The sense of social presence of consciousness mainly affects readers’ feelings of trust and companionship. When recommending books, the app will recommend books based on social connections, and mark the information that “your friends are reading” for the recommended books, to improve the credibility of the recommendation through the influence of friends. After clicking, you can see the reading information of the reading users. The number of reading users has changed from a number to real users or even

some friends, which enhances the user's sense of trust in the system recommendation. In the reading page and operation interface state, you can see the total number of readers of the current book, that is, the number of book friends. Click to see the details of the book friends, their reading progress. This streamlined way of presentation can provide the right sense of conscious presence and bring people a sense of company.

3.3 Emotional Social Presence and Reading Community Experience Elements

Emotional social presence refers to the reader's emotional relationship with others in society and the ability to empathize with others emotionally. Emotional relationships are often generated in behavioral interactions, especially interactive rituals that can stimulate users' personal emotional energy [9]. On the WeRead interface, there is a heart-shaped "like" button after each chapter of a book, which allows users to express their liking for the chapter. Through the display of the number of likes, you can feel the reading mood of other users. All users who like the same chapter have the same focus, i.e., the content of the current paragraph; Readers can share their love of the content, and the interaction is private enough that only those who read the book can participate in it, perfect for an interactive ritual that inspires more intense user emotion. After clicking the "like" button, the interface will not only display the simulation of heartbeat pattern dynamic effect, but also users can feel a short vibration, truly simulate the joy of reading when meeting friends. Vivid and diverse design elements provide conditions for the generation of emotional presence.

Emotional social presence mainly affects readers' emotional experience, social experience, and achievement experience. In the process of praising each other to express the recognition and love of each other's views is itself a kind of emotional transmission. Strong emotional resonance can generate a sense of self-identity, enhance self-confidence, achieve a good emotional experience and gain a sense of belonging to the group which contributes to nice social experience. In addition, the number of likes from others represents the attention and respect of others, expanding personal social influence and feeling self-worth, users can get strong achievement experience.

3.4 Cognitive Social Presence and Reading Community Experience Elements

Cognitive social presence means that in the reading community, users can understand others' views on things and comprehensively judge others' emotional, cognitive, and behavioral tendencies [10]. Cognition is characterized by situational relevance. Situational cognition regards systems -- people, others, society, and natural environment -- as a co-existing whole, which together constitute the vision of cognitive meaning. In the process of reading on WeRead, users can underline their thoughts and see other readers' comments on the current snippet. Multiple readers can post their thoughts on the same clip or reply to or like a comment. Since all comments are based on the same paragraph context, the interactive communication is in a unified and specific reading context, making the user's interaction with the content discussion more targeted and easier to understand. The posting and sharing of readers' thought will also be automatically accompanied by a text location link to the source of the thought, so that visitors can quickly locate the context in which the author post the idea to read. The users in WeRead

are also divided into circles, such as mystery reasoning circles, social observer circles and so on. Circles help users find people with the same interests and similar knowledge background, to better reach tacit understanding in the process of communication, identify with each other's values, carry out smooth exchange of views, and obtain a high sense of cognitive presence.

Cognitive social presence mainly affects readers' emotional experience, thinking experience, achievement experience and social experience. When users find people who share the same values with them, they can "like", "comment" or even "follow" them as friends, quickly establishing intimate psychological connections and feeling like friends at first sight. The published ideas are praised and paid attention by other users. After receiving their attention, users' needs for social recognition are satisfied and achievement experience is achieved. Behaviors such as taking notes, communicating, and sharing, commenting, and forwarding further visualize the emotions and thoughts in reading, deepen the reading experience, and help readers enjoy high-quality reading experience. The expression and communication of ideas enhance the thinking experience.

4 The Construction of an Interaction Design Model

It is concluded that the elements of identity design can increase users' trust in book recommendation and companionship in the reading process, which are mediated by conscious social presence. The elements of interactive design such as interface animation, interactive rituals, can externalize the user's internal emotions, form resonance among users, stimulate users' stronger personal emotional energy to achieve a high emotional presence. The emotional identity that people gain in their interpersonal interactions promotes social influence and a sense of accomplishment. The elements of situational design build a social context in reading which reduces the cognitive cost of users to other people's views, makes it easier for users to understand each other, and gain deep reflection in the high cognitive presence. When each basic experience is built compelling,

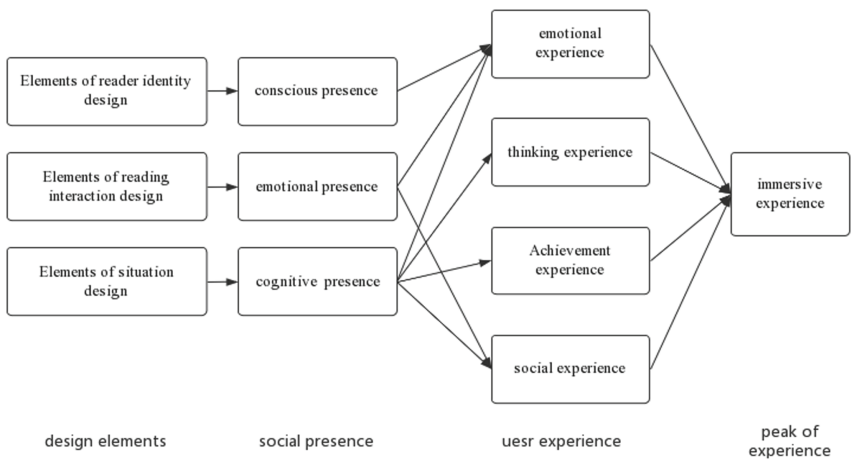


Fig. 1. The experience model of social reading platforms

the immersive experience will be brought into being which is the peak of experience (Fig. 1).

5 Design Strategies for Social Reading Platform

Finally, we put forward reference suggestions for the experience design of future reading communities from the perspective of practical application:

- Focus on the interactive design between user and reading platform, user, and user. Organize interactive groups around reading content and reading interest, design natural and easy interaction, and improve the timeliness and vividness of interaction, to strengthen user engagement.
- Intensify the emotional design of a social reading platform. Using the design method of emotion metaphor build interactive way for users to improve the emotional expression and create emotional resonance between the user, create a pleasant reading atmosphere, to enhance the sense of intimacy and companionship in the reading community, and enhance users' sense of belonging to the community platform.
- Focus on the segmentation of reading and social situations in the design. Before reading, users search for books in a large amount of information and pay attention to experience such as efficiency and trust. When reading, thinking, and accompanying experience are more critical. In the evaluation and expression process after reading, users focus on interaction, social influence, and achievement experience. Design according to specific situations can provide users with more precise services and promote a virtuous cycle of experience in all links.
- Actively apply innovative media that contain more social clues, such as AR technology and video interactive technology, to expand the social presence from a technical perspective, and explore more possibilities of the social reading platform.

6 Conclusion and Prospect

From the perspective of social presence, this paper takes WeRead app as an example to study the mechanism by which specific design elements in the reading community influence user experience through social presence. This study constructs user experience model of online social reading platform. Combined with the future development trend, this paper puts forward constructive suggestions for the experience design of reading community from four aspects: interactive design, emotional design, situational design and combining technological innovation design.

The innovation of this paper is the introduction of the social presence theory to the social reading media. Recognizing the socialized characteristics of reading behavior in the Internet environment, this paper studies users' reading experience from the perspective of socializing social attributes, and innovatively introduces social existence theory, which is a novel research approach.

in the future, Further research is needed in the following aspects. (1) The sample number of user studies is small, and all of them are from China, so its applicability to the global scale needs to be verified. (2) The research on the design of anti-addiction

is limited to qualitative research, and future research can include qualitative research to make the model more accurate.

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